



Advanced Digital Image Processing Lecture #5

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Contents

- ❖ Introduction of Edge detection
- ❖ Point Detection Method
- ❖ Line Detection Method
- ❖ Edge Detection Method
- ❖ Edge Linking and Boundary Detection



Edge Detection (1 of 3)

- ❖ Edges often occur at points where there is a large variation in the luminance values in an image, or occluding boundaries, of the objects in a scene.
- ❖ However, large luminance changes can also correspond to surface markings on objects.
- ❖ Points of tangent discontinuity in the luminance signal (rather than simple discontinuity) can also signal an object boundary in the scene



Edge Detection (2 of 3)

- ❖ The usual approach is to simply define edges as step discontinuities in the image signal.
- ❖ The method of localizing these discontinuities often then becomes one of finding local maxima in the first derivative of the signal, or zero-crossings in the second derivative of the signal.
- ❖ In computer vision, edge detection is traditionally implemented by convolving the signal with some form of linear filter, usually a filter that approximates a first or second derivative operator.



Edge Detection (3 of 3)

Detection of Discontinuity

- ❖ Point Detection
- ❖ Line Detection
- ❖ Edge Detection



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- ❖ Line Detection Method
- ❖ Edge Detection Method
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Point Detection

- ❑ A point has been detected at the location on which the mask is centered if,

$$|R| \geq T$$

- ❑ T is the nonnegative threshold and R,

$$R = w_1 z_1 + w_2 z_2 + \cdots + w_9 z_9$$

- ❑ z_i is the gray level of the pixel associated with the mask coefficient w_i
- ❑ The idea is that isolated point will be quite different from its surroundings

Point Detection Mask

-1	-1	-1
-1	8	-1
-1	-1	-1

- ✓ Sum of all point is Zero
- ✓ Center point is equal to sum of other points but opposite sign



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Line Detection

-1	-1	-1
2	2	2
-1	-1	-1

Horizontal

-1	-1	2
-1	2	-1
2	-1	-1

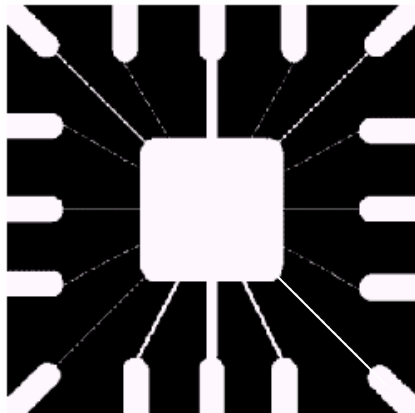
+45°

-1	2	-1
-1	2	-1
-1	2	-1

Vertical

2	-1	-1
-1	2	-1
-1	-1	2

-45°



Binary Wire-bond mask



After processing -45°
line detector



Result of Thresholding



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Edge Detection Method

❖ **First Order Differential Method**

❖ Second Order Differential method



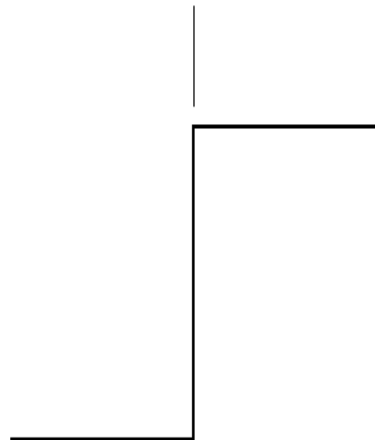
Edge Detection Method

First Order Differential Method

- ❖ Edge occurs where there is a discontinuity in the intensity function or a very steep intensity gradient in the image
- ❖ Using this assumption, if we take the derivative of the intensity values across the image and points where the derivative is a maximum, we will have marked our edges

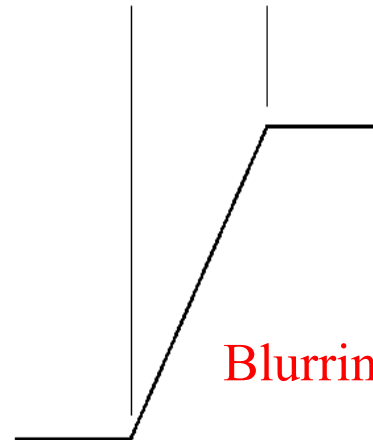
Model of a Digital Edge

Model of an ideal digital edge



Gray-level profile
of a horizontal line
through the image

Model of a ramp digital edge



Gray-level profile
of a horizontal line
through the image

Blurring in the edge

a b

FIGURE 10.5

(a) Model of an ideal digital edge.
(b) Model of a ramp edge. The slope of the ramp is proportional to the degree of blurring in the edge.

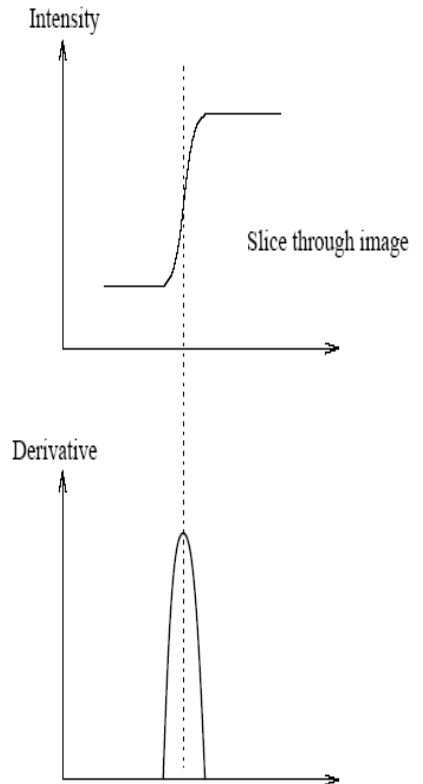
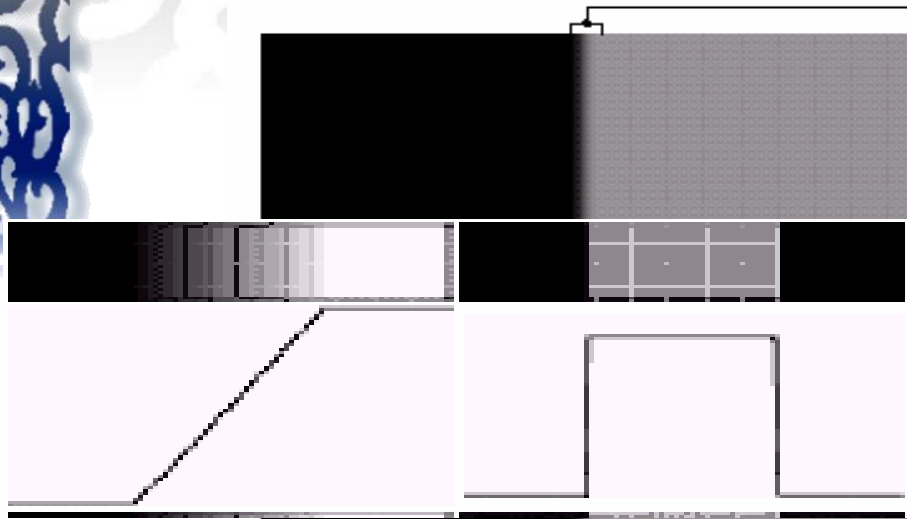
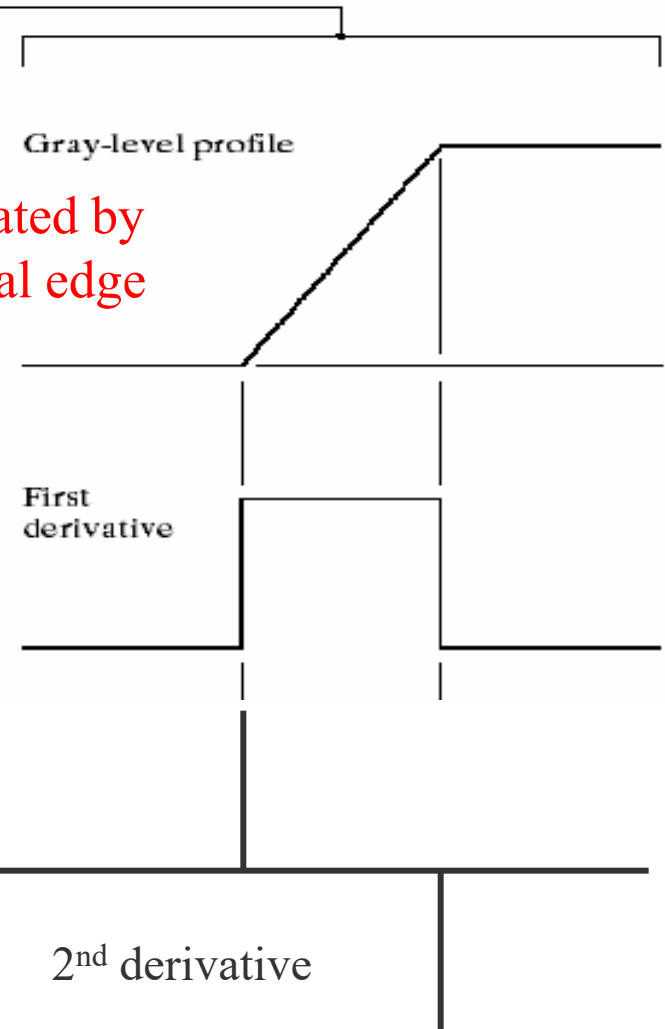


Figure 1: An ideal step edge and its derivative profile.

Model of First Derivative Edge Detection



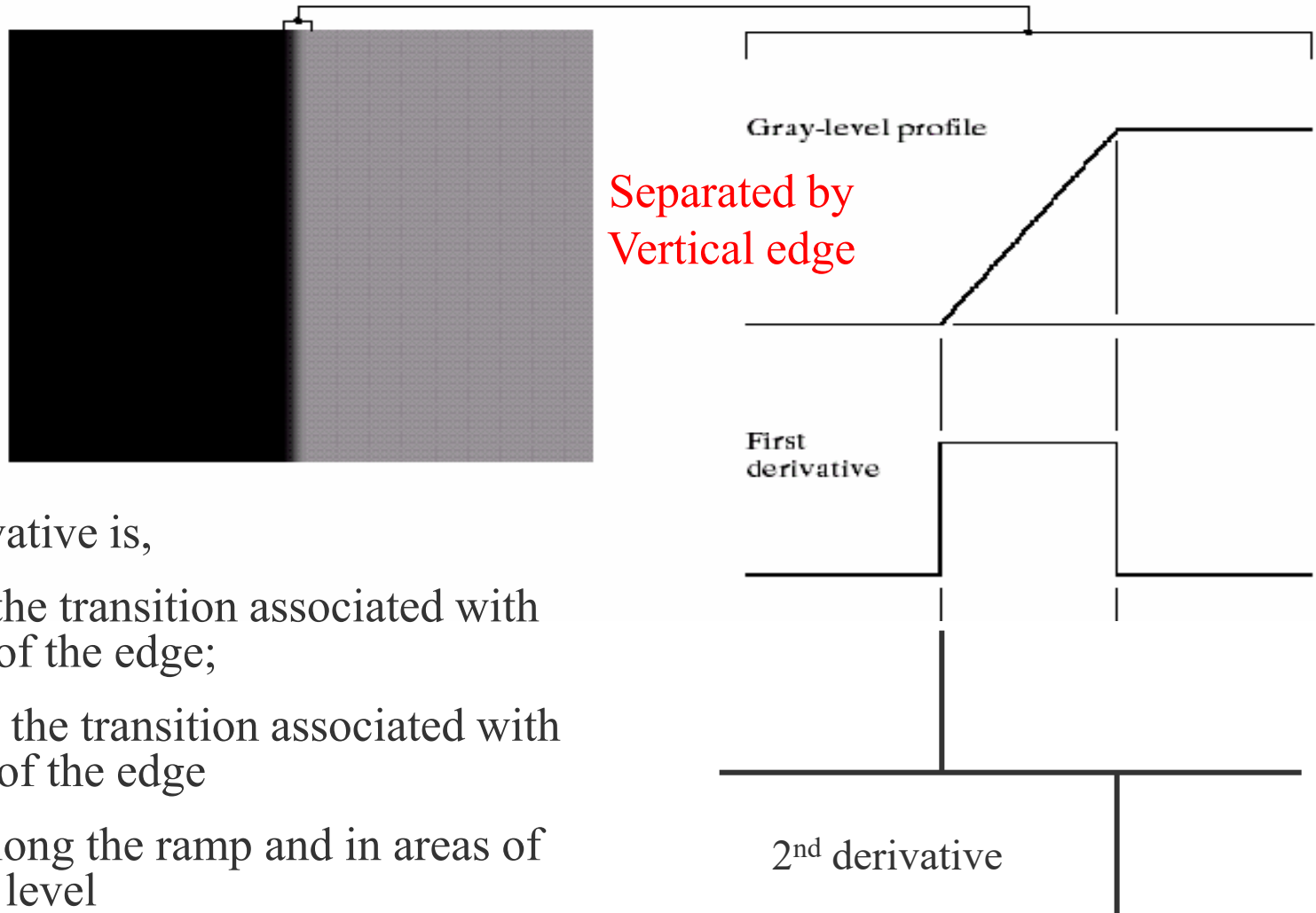
Separated by
Vertical edge



The first derivative is,

- ❑ positive at the point of transition into and out of the ramp as we move from left to right along the profile;
- ❑ Constant for points in the ramp
- ❑ And zero in areas of constant gray level

Model of First and Second Derivative Edge Detection



The 2nd derivative is,

- ❑ positive at the transition associated with the dark side of the edge;
- ❑ Negative at the transition associated with the light side of the edge
- ❑ And zero along the ramp and in areas of constant gray level

Applications of First and Second Derivatives of an Image

- ❖ The Magnitude of the first derivative can be used to detect the presence of an edge at a point in an image
- ❖ The sign of the second derivatives can be used to determine whether an edge pixel lies on the dark and light side of an edge
- ❖ An imaginary straight line joining the extreme positive and negative values of the second derivative would cross zero near the midpoint of the edge.

(Zero Crossing property)

First Order derivative (Gradient Operators)

- ❖ First order derivatives of a digital Image are based on various approximations of the 2-D gradient,
- ❖ The gradient of an image $f(x,y)$ is defined as the vector,

$$\overline{\nabla f} = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}, \quad \text{and magnitude}$$

$$\nabla f = \text{mag}(\overline{\nabla f}) = [G_x^2 + G_y^2]^{1/2}$$

$$\Delta\lambda \approx |\mathbf{C}^x| + |\mathbf{C}^y|$$

Example of Edge Detection (1)

a
b c
d e
f g

FIGURE 10.8

A 3×3 region of an image (the z 's are gray-level values) and various masks used to compute the gradient at point labeled z_5 .

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

-1	0	0	-1
0	1	1	0

Roberts

-1	-1	-1	-1	0	1
0	0	0	-1	0	1
1	1	1	-1	0	1

Prewitt

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

Sobel

First order partial derivatives at point z_5

$$G_x = (z_9 - z_5)$$

$$G_y = (z_8 - z_6)$$

**Roberts Gradient Operator: No
clear center**

Example of Edge Detection (2)

a
b c
d e
f g

FIGURE 10.8

A 3×3 region of an image (the z 's are gray-level values) and various masks used to compute the gradient at point labeled z_5 .

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

-1	0	0	-1
0	1	1	0

Roberts

-1	-1	-1	-1	0	1
0	0	0	-1	0	1
1	1	1	-1	0	1

Prewitt

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

Sobel

First order partial derivatives at point z_5

$$G_x = (z_7 + z_8 + z_9) - (z_1 + z_2 + z_3)$$

$$G_y = (z_3 + z_6 + z_9) - (z_1 + z_4 + z_7)$$

Prewitt Gradient Operator: differences between
1st and 3rd rows x direction;
3rd and 1st columns y direction

Example of Edge Detection (3)

a
b c
d e
f g

FIGURE 10.8
A 3×3 region of an image (the z 's are gray-level values) and various masks used to compute the gradient at point labeled z_5 .

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

-1	0	0	-1
0	1	1	0

Roberts

-1	-1	-1	-1	0	1
0	0	0	-1	0	1
1	1	1	-1	0	1

Prewitt

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

Sobel

First order partial derivatives at point z_5

$$G_x = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$$

$$G_y = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$$

Sobel Gradient Operator: A weight value of 2 is used in the center point to achieve some smoothing

Prewitt and Sobel Masks for Detecting Diagonal Edges

0	1	1
-1	0	1
-1	-1	0

-1	-1	0
-1	0	1
0	1	1

Prewitt

0	1	2
-1	0	1
-2	-1	0

-2	-1	0
-1	0	1
0	1	2

Sobel

Edge Detection

Second-order Derivative

- ❖ The Laplacian of a 2-D function $f(x,y)$ is a second-order derivative defined as,

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Second-order Derivative Laplacian Mask

Z_1	Z_2	Z_3
Z_4	Z_5	Z_6
Z_7	Z_8	Z_9

0	-1	0
-1	4	-1
0	-1	0

-1	-1	-1
-1	8	-1
-1	-1	-1

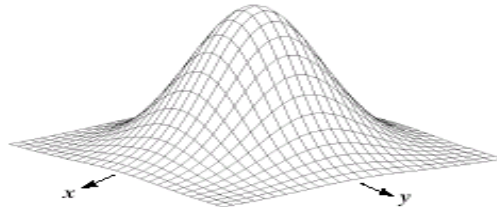
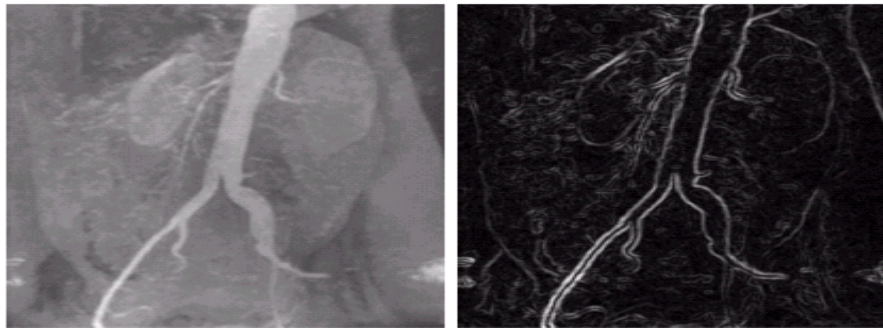
$$\nabla^2 f = 4z_5 - (z_2 + z_4 + z_6 + z_8)$$

$$\nabla^2 f = 8z_5 - (z_1 + z_2 + z_3 + z_4 + z_5 + z_6 + z_7 + z_8)$$

Diagonal Neighbors

Example: Edge Detection

(a) Original Image, (b) Sobel Gradient,
(c) Spatial Gaussian Smoothing Function, (d) Laplacian Mask,
(e) Log, (f) Threshold, (g) Zero Crossing



-1	-1	-1
-1	8	-1
-1	-1	-1



EDGE Operator in Matlab (1)

- ❖ EDGE takes an intensity image $I(x,y)$ as its input, and returns a binary image $BI(x,y)$ of the same size as I , with 1's where the function finds edges in $I(x,y)$ and 0's elsewhere.
- ❖ EDGE supports six different edge-finding methods:
 - ❖ The Sobel method finds edges using the Sobel approximation to the derivative. It returns edges at those points where the gradient of $I(x,y)$ is maximum.
 - ❖ The Prewitt method finds edges using the Prewitt approximation to the derivative. It returns edges at those points where the gradient of $I(x,y)$ is maximum.
 - ❖ The Roberts method finds edges using the Roberts approximation to the derivative. It returns edges at those points where the gradient of $I(x,y)$ is maximum.

EDGE Operator in Matlab (2)

- ❖ The Laplacian of Gaussian method finds edges by looking for zero crossings after filtering I with a Laplacian of Gaussian filter.
- ❖ The zero-cross method finds edges by looking for zero crossings after filtering I with a filter you specify.
- ❖ The Canny method finds edges by looking for local maxima of the gradient of I . The gradient is calculated using the derivative of a Gaussian filter. The method uses two thresholds, to detect strong and weak edges, and includes the weak edges in the output only if they are connected to strong edges. This method is therefore less likely than the others to be "fooled" by noise, and more likely to detect true weak edges.

Example of Edge Detection



Original,

Sobel,

Sobel-horizontal, Vertical



prewitt,

roberts,

canny,

Zero cross



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Edge Linking and Boundary Detection



Boundary/Contour



Edges





Edge Linking and Boundary Detection (2)

- ❖ **Edge Linking procedures** assemble edge pixels into meaningful pixels
- ❖ For linking edge points, we need to analyze the characteristics of pixels in a small neighborhood (3×3 or 5×5) about every point (x, y) in an image that has been labeled an edge point
- ❖ All points that are similar according to a set of predefined criteria are linked

How to Establish Similarity of Edge Pixels?

- ❖ Two principal properties used for establishing similarity of edge pixels,
 - ❑ The strength of the response of the gradient operator (∇f) used to produce the edge pixel, and
 - ❑ The direction of the gradient vector ($\alpha(x,y)$)
- ❖ A point in the predefined neighborhood of (x, y) is linked to the pixel at (x,y) if both magnitude and direction criteria are satisfied



List of Reference Books

- ❖ Rafael C. Gonzalez, “Digital Image Processing”.
- ❖ Anil K. Jain, “Fundamental of Digital Image Processing”.
- ❖ Dana H. Ballard, “Computer Vision”
- ❖ David H. Hubel, “Eye, Brain, and Vision”
- ❖ Dwayne Phillips, “Image Processing in C”



End of Today's Lecture

Questions?

Thanks to all!